Some useful definitions

EM_bright classification:

- <u>P(HasNS)</u> —> at least one of the compact objects in the binary has a mass that is consistent with a neutron star —> 1 M_sun < M < **M_max (EoS)**
- <u>P(HasRemnant)</u> —> Non-zero mass is left around the final compact object. Depends on the mass ratio, priors on spins, EoS —> each pipeline can give different values
- <u>P(HasMassGap)</u> —> one object in the 3-5 M_sun range

Coarse-grained Mass Information (CGMI):

- An array of chirp mass bins from [0.1, … , 1000] M_sun with associated binned probability [P(n,n+1)]

$$\mathcal{M}_{\text{chirp}} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}$$

Digression on p_astro and EM_bright properties